Exact Normalization Windows and Sharp Mixed-Norm Tails for the Source Backward Intertwiner

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Abstract

The source constructs a measurable family of bilateral shifts over the 2-adic adding machine and normalizes a backward intertwiner by a countable supremum. Its pointwise drift constant is not integrable and does not provide a uniform truncation bound. We exploit the actual nested exceptional residues: after a reference depth $8^{v_2(x+1)}$, the accumulated exceptional drift cancels exactly. On a cylinder set of measure $1-2^{-R-1}$, the normalizing supremum is therefore an exact maximum over $|j|<\max(64,2\cdot 8^R)$, determined by finitely many residue digits. For coordinate cutoffs at depth M, this gives the sharp order $M^{-(1/2-1/p)/3}$ from L^p of the base with values in ℓ^2 to L^2, for every $2<p\le\infty$. Rare cylinders prove that the same tails have L^2 operator norm exactly one. We construct finite coefficient-table approximants with exponentially small commutators and explain why these are neither finite-rank operators nor substitutes for the source's exact commutant argument. A forward resolvent enclosure is recorded separately. Weighted translation classification, companion decomposition lemmas, and the source global hyperinvariant-subspace conclusion are not independently certified here. Contribution assessment: C3/5, exact source-specific normalization windows and matching mixed-norm truncation bounds provide a notable local refinement without a new global invariant-subspace theorem.

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1 The model and the source proof boundaries

All Hilbert spaces are complex. Let $X = \mathbb{Z}_2$ have completed Haar probability μ, let $\theta x = x + 1$, and set

$$K = \ell^2(\mathbb{Z}), \qquad H = L^2(X, \mu; K), \qquad H_p = L^p(X, \mu; K) \quad (2 \le p \le \infty).$$

The norm in H_p is the norm of $x \mapsto \|f(x)\|_K$, not the coordinatewise ℓ^p norm. Write $K_{\ge h}$ for the closed span of e_j, $j \ge h$, and Π_M for the orthogonal projection onto coordinates $|j| < M$, acting fiberwise. All identities on X are modulo null sets unless stated otherwise. Probability preservation of θ is essential to every global norm below.

The source [1] asserts a nonzero norm-quasinilpotent bounded operator S with no nonzero proper closed *hyperinvariant* subspace: the subspace must be invariant under every bounded operator commuting with S. This is not the assertion that S has no ordinary invariant subspace. In fact, $L^2(X; K_{\ge h})$ is S-invariant for every integer h. The claimed commutant is a proper strongly closed unital transitive algebra. These are source claims, not consequences of the finite checks in this report.

1.1 Actual weights and the forward operator

Put $c = 10$, $D = 100$, $N_m = 2^m$, $I_m = [8^{m-1}, 8^m) \cap \mathbb{Z}$, and $T_m(x) = x \bmod 2^m$. Set $J_m = D(2^m - 2)$ and $a_0^{\pm} = c$. For $k \in I_m$ define

$$\begin{aligned} a_k^+(x) &= c(k+1) + J_m + DT_m(x), \\ a_k^-(x) &= c(k+1) + J_m + D(2^m - 1 - T_m(x)). \end{aligned} \tag{1}$$

The bilateral coefficients and operators are

$$a_j = \begin{cases} a_j^+ & j \ge 0, \\ a_{-j-1}^- & j < 0, \end{cases} \qquad \beta_j = e^{-a_j}, \qquad S_x e_j = \beta_j(x) e_{j+1}, \qquad (Sf)(x) = S_x f(x).$$

In particular $\beta_{-1} = \beta_0 = e^{-10}$. Directly from the formulas,

$$0 \le a_k^{\pm} - c(k+1) \le 400 k^{1/3} \quad (k \ge 1). \tag{2}$$

Indeed $2^m \le 2k^{1/3}$ on I_m, and the nonnegative correction is at most $2D2^m$. The source proves

$$\|S\| = e^{-c}, \qquad \|S^n\| \le e^{-cA(n)}, \qquad A(n) = \left\lfloor \frac{(n+1)^2}{4} \right\rfloor \quad (n \ge 0). \tag{3}$$

For completeness, $a_j \ge cd(j)$, with $d(j) = j + 1$ for $j \ge 0$ and $d(j) = -j$ otherwise. The minimum sum of n consecutive $d(j)$ is $m(m+1)$ for $n = 2m$ and $(m+1)^2$ for $n = 2m+1$. Taking the supremum of weighted-shift products proves (3), including $A(0) = 0$. This is a retained source estimate, not a new result.

1.2 Architecture, including the imported steps

The entire cached scientific manuscript was literally read. Its chain is more substantial than a finite shift computation:

1. `criterion.tex` and the included `domar-proof.tex` give a quadratic weighted-translation classification. Decreasing weights, one-sided convexity/concavity of logarithms, summable root tails, and quadratic constant $a > \log 3$ enter a contact-point convolution argument. This is the source's specialization of classical weighted-translation theory [2].

2. `weights.tex` verifies those conditions fiberwise, with $\log w_{\pm n} = \mp 5n(n+1) + O(n^{4/3})$, uniformly in the base, and proves the forward power estimate.

3. `intertwiner.tex` constructs the exact normalized V. The local results below use only its explicit recurrence and weights.

4. `subspaces.tex` uses the companion projection-field lemma to represent one fixed hyperinvariant subspace by measurable fiber subspaces, applies the fiber classification, and uses V to force a strictly descending integer tail label under θ. Probability preservation and ergodicity rule out finite labels.

5. `conclusion.tex` explains noncompactness, infinite multiplier multiplicity, the polar algebra, and the difference between finite atomic joint data and the diffuse algebra of the full family.

The companion field lemma and commutant lemma are imported; their bodies were not freshly read. Reading the source's Domar proof does not constitute an audit of all historical theorems it references. In the measurable-subspace argument the null set is chosen after one fixed subspace. There is no intersection over all subspaces. None of our estimates removes this quantifier or replaces the source's exact classification step.

2 The exact drift and its finite reference depth

Let X_* be the source's invariant conull set. For $x \in X_*$ put $r(x) = v_2(x+1)$; it is finite. The exceptional residue $T_m(x) = 2^m - 1$ occurs exactly when $m \le r(x)$. The exceptional events are nested, not independent. For $r = 0, 1, \ldots,$

$$\mu\{r(x) = r\} = 2^{-r-1}, \qquad E_R = \{r(x) \le R\}, \qquad \mu(E_R^c) = 2^{-R-1}. \tag{4}$$

Intersecting these sets with X_* does not change their measures. The possible integer representatives used to describe cylinders are not asserted to belong to X_*.

The source defines $\widetilde{b}_0 = 1$ and

$$\widetilde{b}_{j+1}(x) = \widetilde{b}_j(x)\frac{\beta_{j-1}(\theta x)}{\beta_j(x)}, \qquad L(x) = \sup_{j\in\mathbb{Z}} \widetilde{b}_j(x), \qquad b_j = \widetilde{b}_j/L. \tag{5}$$

On X_* both tails of $\widetilde{b}_j$ tend to zero, so L is a finite *attained* positive maximum. Off X_* set $b_j = 0$. Then $V_x e_j = b_j(x) e_{j-1}$ and

$$(Vf)(\theta x) = V_x f(x), \qquad \|V\| \le 1, \qquad SV = VS. \tag{6}$$

Every coefficient is positive on X_*, giving dense images of the corresponding tails. Because $b_j \to 0$ at both ends, V_x is compact and not onto K; positivity does not give a bounded inverse.

Define the partial exceptional sum

$$F_r(n) = \sum_{m=1}^{r} 2^m |I_m \cap [1, n)|, \qquad F_r = \frac{14}{15}(16^r - 1), \qquad h_r = 8^r.$$

Here $F_0 = 0$. For $n \ge h_r$, $F_r(n) = F_r$, since $2^m|I_m| = (7/8)16^m$ and all exceptional blocks have ended. Taking logarithms in (5), write $\ell_j(x) = \log \widetilde{b}_j(x)$. The exact formulas are

$$\begin{aligned} \ell_n &= -D(n-1) + DF_r(n) + a_{n-1}^+(\theta x) - c, && n \ge 1, \\ \ell_{-q} &= -D(q-1) + DF_r(q) + a_q^-(\theta x) - c, && q \ge 1. \end{aligned} \tag{7}$$

The negative boundary term is a_q^-, not a_{q-1}^-. To verify these expressions, sum $\ell_{j+1} - \ell_j = a_j(x) - a_{j-1}(\theta x)$. The forward drift is $D(n-1) - DF_r(n)$; the drift on indices $-q, \dots, -1$ is its negative with $n = q$. This also checks the sign in the second formula.

Remark 2.1 (Why the unnormalized constant is insufficient)**.** The source's pointwise drift allowance can be chosen as

$$C(x) = D(1 + F_{r(x)}) = \frac{20}{3}(1 + 14 \cdot 16^{r(x)}).$$

Its positive p-th moment is finite exactly for $p < 1/4$, by (4). In particular it is not integrable. Substitution of this allowance into an exponential tail estimate cannot produce a uniform-in-x or integrable majorant for the unnormalized sequence. The cancellation below uses the normalization rather than integrating that allowance.

Lemma 2.2 (Cancellation after the exceptional depth)**.** *For $x \in X_*$, $r = r(x)$, and $h = 8^r$, the following bounds hold:*

$$\begin{aligned} \ell_n - \ell_h &\le -90(n-h) + 400n^{1/3} && (n \ge h), \\ \ell_{-q} - \ell_{-h} &\le -90(q-h) + 400q^{1/3} && (q \ge h). \end{aligned} \tag{8}$$

Consequently each right side bounds $\log b_n$ or $\log b_{-q}$, respectively, and $b_j \le 1$ as well.

Proof. In the first difference of (7), the terms DF_r cancel. The difference of the remaining boundary weights is at most $c(n-h) + 400(n-1)^{1/3}$ by (2) and the nonnegative correction at the reference index. This proves the first bound, also for $h = 1$, where the reference correction is zero. For the negative difference the boundary weights are a_q^- and a_h^-; the same argument gives $c(q-h) + 400q^{1/3}$. Finally $L \ge \widetilde{b}_h, \widetilde{b}_{-h}$. □

3 A finite exact normalizer on large cylinder sets

Theorem 3.1 (Finite normalization certificate)**.** *For every integer $R \ge 0$ put*

$$M_R = \max(64, 2 \cdot 8^R), \qquad d_R = \max(2, R+1).$$

For almost every $x \in E_R$,

$$L(x) = \max_{|j| < M_R} \widetilde{b}_j(x), \qquad b_j(x) \le e^{-20|j|} \quad (|j| \ge M_R). \tag{9}$$

The maximum and all coefficients b_j, $|j| < M_R$, are functions of the single finite residue $T_{d_R}(x)$ on E_R.

Proof. For $s = |j| \ge M_R$, we have $h_r \le s/2$ and $s \ge 64$. Hence $400s^{1/3} \le 25s$, and (8) is at most $-45s + 25s = -20s$. In particular every excluded positive coefficient is strictly below $\widetilde{b}_{h_r}$ and every excluded negative coefficient is strictly below $\widetilde{b}_{-h_r}$. These reference indices belong to the retained window. This proves the maximum identity, not just an approximation to it.

The recurrence for $|j| < M_R$ involves only weights with $k \le M_R - 1$. For $R = 0, 1$, these indices are below $64 = 8^2$; for $R \ge 2$, they are below $2 \cdot 8^R < 8^{R+1}$. Thus the weights depend only on $T_{d_R}(x)$ and the corresponding residue of $x + 1$. The event E_R itself is determined by $T_{R+1}(x) \ne 2^{R+1} - 1$. All assertions therefore hold cylinderwise modulo the fixed null set. □

The coefficient table is particularly simple in logarithmic form. For each good residue, start at $\ell_0 = 0$, use integer additions from (5), subtract the largest retained integer logarithm, and exponentiate. This avoids constructing the potentially huge number L before dividing. An interval implementation can enclose the final exponentials after the exact integer stage. No floating evaluation or rounded comparison is needed to locate the largest integer logarithm.

Example 3.2 (A completely specified cylinder)**.** On $T_2(x) = 0$ we have $r = 0$. For $1 \le k < 8$, the positive weight correction at x is zero and that at θx is 100; the negative corrections have the opposite values. Formula (7) gives $\ell_1 = 0$, $\ell_2 = 10$, and $\ell_{-1} = 10$. For $2 \le n \le 8$, $\ell_n = 190 - 90n$, while for $1 \le q < 8$, $\ell_{-q} = 100 - 90q$. For $8 \le k < 64$, the corrections at θx are 300 on the positive side and 400 on the negative side. Thus $\ell_9 = -420$ and $\ell_{-8} = -220$, and the subsequent values decrease by 90 inside this block. Together with Theorem 3.1, this gives $\log L = 10$, attained at $j = -1, 2$. In particular $b_0 = b_1 = e^{-10}$, not one. These are statements on the full cylinder outside a null set, not evaluations at the exceptional embedded integer $x = 0$ itself.

4 Sharp mixed-norm tails and the operator-norm obstruction

Theorem 4.1 (Two-sided truncation order)**.** *Let $2 < p \le \infty$, put $a_p = 1/2 - 1/p$, and let $M \ge 64$ be an integer. Then*

$$4^{-a_p}(M+2)^{-a_p/3} \le \|V(I - \Pi_M)\|_{H_p \to H} \le e^{-20M} + (2/M)^{a_p/3}. \tag{10}$$

For every integer $M \ge 1$,

$$\|V(I - \Pi_M)\|_{H \to H} = 1. \tag{11}$$

Nevertheless $V\Pi_M f \to Vf$ in H for each fixed $f \in H$. The exponent $a_p/3$, in particular $1/6$ for $p = \infty$, is thus optimal for these coordinate cutoffs. No assertion of optimality over other approximation schemes is made.

Upper bound. Choose $R = \lfloor \log_8(M/2) \rfloor$, which is nonnegative. On E_R we have $h_r \le M/2$; the proof of Theorem 3.1 gives $\sup_{|j| \ge M} b_j(x) \le e^{-20M}$. On E_R^c the same supremum is at most one. Probability preservation of θ gives

$$\|V(I - \Pi_M)f\|_H \le e^{-20M} \|f\|_H + \left\|\mathbf{1}_{E_R^c} f\right\|_H \le \left(e^{-20M} + 2^{-(R+1)a_p}\right) \|f\|_{H_p}.$$

The second inequality is Holder's inequality on the probability base, including $p = \infty$. Since $R + 1 > \log_8(M/2)$, $2^{-R-1} < (2/M)^{1/3}$. This proves the upper bound. □

Lemma 4.2 (A maximizer must escape on a rare cylinder)**.** *If $r(x) = r \ge 1$ and $h = 8^r$, the sequences*

$$\ell_1, \ell_2, \dots, \ell_h \quad \text{and} \quad \ell_0, \ell_{-1}, \dots, \ell_{-(h-1)}$$

are strictly increasing, and $\ell_1 = \ell_0 = 0$. If $h \ge M + 2$, no maximizing index for $\widetilde{b}_j$ lies in $|j| < M$.

Proof. For $1 \le k < h$, the block containing k has level $m \le r$. Thus $T_m(x) = 2^m - 1$ and $T_m(\theta x) = 0$. The positive correction in $a_k^+(x)$ and the negative correction in $a_k^-(\theta x)$ both equal $D(2 \cdot 2^m - 3)$. The opposite two corrections at that level equal $D(2^m - 2)$. For $k = 1$, comparison with the central weight gives an increment $c + D > 0$. For $k > 1$, let m' be the level of $k - 1$. It is either m or $m - 1$. Both differences

$$a_k^+(x) - a_{k-1}^+(\theta x), \qquad a_k^-(\theta x) - a_{k-1}^-(x)$$

equal $c + D(2 \cdot 2^m - 2^{m'} - 1) > 0$. They are respectively $\ell_{k+1} - \ell_k$ and $\ell_{-k} - \ell_{-(k-1)}$. The central equality follows from $a_0 = a_{-1} = c$. Each positive index in the retained window is strictly beaten by h, and each negative index by $-(h-1)$; index zero is beaten as well. The strict inequalities prove the last claim. □

Lower bound and norm assertion in Theorem 4.1. Set $r = \lceil \log_8(M+2) \rceil$ and $F = \{r(x) = r\} \cap X_*$. Then

$$\mu(F) = 2^{-r-1} \geq \frac{1}{4(M+2)^{1/3}}.$$

For each $x \in F$, choose the first maximizing index in a fixed enumeration of $\mathbb{Z}$. Such a choice $j(x)$ is measurable, since the $\widetilde{b}_j$ and their attained countable supremum are measurable. Lemma 4.2 ensures $|j(x)| \geq M$ and $b_{j(x)}(x) = 1$. The strongly measurable field

$$f(x) = \mu(F)^{-1/p} \mathbf{1}_F(x) e_{j(x)}$$

has H_p norm one and output norm $\mu(F)^{a_p}$; the convention for $p = \infty$ is $\mu(F)^0 = 1$ in the input factor. This proves the lower bound. With $p = 2$, it proves norm at least one, while the contraction property proves the reverse inequality. Finally, $\Pi_M \to I$ strongly on H, so boundedness of V proves strong convergence without a uniform rate on the H unit ball. □

This obstruction is specific: the optimizing input depends on M and is concentrated in a progressively rarer base cylinder with its coordinate maximum moved beyond the cutoff. Neither pointwise decay of the coefficients nor strong convergence controls these moving inputs. It is not an assertion that any fixed input fails to converge.

5 Finite coefficient tables with certified commutator error

Theorem 5.1 (Table approximants)**.** *Define*

$$W_R = V M_{\mathbf{1}_{E_R}} \Pi_{M_R},$$

where M_g denotes scalar multiplication on the input base. Then W_R is a contraction, its nonzero coefficient columns are exactly computable from a finite residue table, and for $2 < p \leq \infty$,

$$\|V - W_R\|_{H_p \to H} \leq e^{-20M_R} + 2^{-(R+1)a_p}, \tag{12}$$

$$\|[S, W_R]\|_{H \to H} \leq e^{-10M_R}. \tag{13}$$

It converges strongly to V, but $\|V - W_R\|_{H \to H} = 1$ for every finite R.

Proof. The input operator $M_{\mathbf{1}_{E_R}} \Pi_{M_R}$ is an orthogonal projection. The finite table statement is exactly Theorem 3.1, with zero coefficients on bad cylinders. Splitting the input over E_R and E_R^c proves (12). Pointwise eventual membership in E_R, coordinate convergence and dominated convergence prove the strong assertion. On the positive measure set E_R^c, V_x has attained norm one. The measurable maximizer construction in the preceding proof, now supported there, gives $\|V - W_R\| \geq 1$. The reverse inequality follows because $V - W_R$ is V times an orthogonal projection.

Since S commutes with V and with every scalar multiplier,

$$[S, W_R] = V M_{\mathbf{1}_{E_R}} [S, \Pi_{M_R}].$$

For Π_M, the only nonzero input columns of the latter commutator are $j = M - 1$ (leaving the window) and $j = -M$ (entering it). Their output columns are distinct. Both coefficients have magnitude at most e^{-10M}, by the linear part of the actual weight formulas. Taking the essential supremum proves (13). □

5.1 What a finite table does not mean

Each residue cylinder still supports an infinite-dimensional L^2 space. A table of finitely many coefficient values is therefore not a finite-rank operator on H. For example, b_0 has a fixed positive value on a good cylinder of W_R. An orthonormal sequence of scalar functions on that cylinder, multiplied by e_0, is mapped by W_R to an orthogonal sequence of the same positive norm at coordinate -1 over the translated cylinder. Thus W_R is noncompact. The same argument applies to V.

For comparison, the forward weights give

$$\|S - \Pi_M S \Pi_M\| \le e^{-10M}. \tag{14}$$

The error is a weighted shift whose omitted columns include $-M$, $M-1$ and the exterior tails; all obey this bound. Nonetheless S itself is noncompact: for any scalar orthonormal sequence φ_n in $L^2(X)$, $S(\varphi_n e_0) = e^{-10}\varphi_n e_1$. Hence true finite-rank spatial approximants cannot converge in operator norm to S, V, or W_R. Fiber-coordinate truncation is a different operation.

Every W_R has a nontrivial kernel, including all inputs on bad cylinders. Small commutator norm alone supplies neither exact commutation nor the dense backward-tail images required in the source subspace argument. In particular (13) cannot be inserted into that exact proof as though W_R belonged to the commutant.

5.2 Accuracy, table size, and a cost limitation

There are 2^{d_R} possible residues and at most $2M_R - 1$ retained input columns. Each row can be computed with $O(M_R)$ integer additions after the weights are generated. Thus a straightforward table uses $O(2^{d_R} M_R) = O(16^R)$ entries for large R. This is a count of entries and arithmetic recurrences, not a measured running time or bit-complexity improvement. The exact integers grow with R, and rigorous exponential enclosures have a separate cost. For $H_\infty \to H$ accuracy of order ε, the rare-set term in (12) requires R of order $2\log_2(1/\varepsilon)$ and a window of order ε^{-6}. The certificate is explicit but can be expensive.

For example, $R = 3$ gives $M_R = 1024$, $d_R = 4$, and good measure $15/16$. The error bound is $1/4 + e^{-20480}$ for $p = \infty$, while the commutator bound is e^{-10240}. These very different numbers show why a small commutator is a poor proxy for accurate action on all inputs. Increasing R improves the mixed-norm error, but never the $H \to H$ distance one.

6 A secondary forward resolvent enclosure

The source's forward power estimate also permits an elementary all-radius enclosure. This is a quantitative use of a classical Neumann series, not the principal contribution or a spectral classification theorem.

Proposition 6.1 (Tail and perturbation budgets)**.** *Let $z \neq 0$, $r = |z|$, and $t_n = r^{-n-1}e^{-cA(n)}$. Then*

$$(zI - S)^{-1} = \sum_{n=0}^{\infty} z^{-n-1} S^n$$

converges in operator norm. If $q_N = r^{-1}\exp[-c\lfloor (N+3)/2 \rfloor] < 1$, its tail after the term $n = N$ has norm at most $t_{N+1}/(1 - q_N)$. Moreover

$$\left\|(zI - S)^{-1}\right\| \le B(r) := r^{-1}\left(1 + 2\sqrt{\pi/c}\right)\exp\left(\frac{\max(\log(1/r) - c/2, 0)^2}{c}\right). \tag{15}$$

For any bounded perturbation E with $\|E\|\, B(r) < 1$, the point z belongs to the resolvent of $S + E$, with inverse norm at most $B(r)/(1 - \|E\|\, B(r))$.

Proof. The ratio of successive positive majorants is $t_{n+1}/t_n = r^{-1}e^{-c\lfloor (n+2)/2\rfloor}$, nonincreasing and tending to zero. This proves convergence and the geometric tail bound. Multiplication of the convergent series by $zI - S$ telescopes on either side, so it is the two-sided inverse.

For all $n \geq 0$, $A(n) \geq n^2/4 + n/2$. Write $b = \log(1/r) - c/2$. If $b > 0$, completing the square bounds the sum by

$$r^{-1}e^{b^2/c} \sum_{n\geq 0} \exp\left[-\frac{c}{4}(n - 2b/c)^2\right].$$

For every real center u, a unimodal nonnegative function has its integer sample sum bounded by its maximum plus its integral over the real line: remove one sample adjacent to the maximum and compare the samples on the two remaining monotone tails to disjoint unit intervals. Applied to $e^{-c(t-u)^2/4}$ this gives $1 + 2\sqrt{\pi/c}$. If $b \leq 0$, drop bn and apply the same bound with center zero. This proves (15). Finally factor $zI - S - E = (zI - S)(I - (zI - S)^{-1}E)$ and apply the norm-convergent geometric inverse. This perturbation statement does not preserve hyperinvariance or the source commutant structure. □

7 Original Ram comparisons and the correct state space

Three original private chapter bodies were freshly read for this report, not inferred from catalogue markers or earlier comparisons. Their contribution is methodological; the new bounds above are proved from the assigned source's actual weights.

The state/domain chapter [6] requires a bounded map and bounded inverse between the declared spaces before transporting a bounded-operator or stability statement. Here the multiplication map

$$U_w : \ell^2(w) \longrightarrow \ell^2, \qquad (U_w z)_j = w_j z_j, \qquad \|z\|^2_{\ell^2(w)} = \sum_j |w_j z_j|^2$$

is a unitary between these two specified spaces, and $U_w R U_w^{-1} e_j = (w_{j+1}/w_j)e_{j+1}$. It is not a bounded similarity on unweighted ℓ^2: w_j tends to zero at one end and infinity at the other. Likewise the normalization in (5) does not produce an inverse for V. This distinction prevents a false transport of spectral or invariant subspace statements.

The original spectral-limits chapter [7] separates strong convergence on each fixed vector from norm convergence, and requires control of omitted directions in finite-section arguments. Theorem 4.1 makes that distinction quantitative in the present model: the escaping test vector moves in both its base support and its fiber coordinate. A fixed-vector test can miss it. This is more than renaming a finite Gram matrix, but it is not a new general principle about strong convergence.

The original weighted-sequence chapter [8] proves weighted ℓ^1 convolution bounds and finite-plus-tail residual estimates. We use the discipline of a retained finite part plus a proved infinite complement, not its Banach-algebra conclusions: H_p here has an ℓ^2 fiber, the base is nonatomic, and no weighted ℓ^1 inverse theorem supplies V^{-1}. The fixed-point result referenced by that chapter and other referenced chapter bodies were not read in this assignment and are not invoked.

8 Comparison, current status, and limitations

The source already proves pointwise decay of the unnormalized backward coefficients, bounded exact commutation after normalization, dense backward-tail images, and a uniform forward power bound. It also explicitly warns that its drift allowance can be unbounded and nonintegrable. We retain those statements rather than identifying a gap where the source makes no uniform assertion.

The increment is the cancellation of the accumulated exceptional sum against an actual reference coefficient, the resulting exact finite normalizer on quantified cylinders, and the matching mixed-norm coordinate-tail order. The finite-table commutator certificate and the explicit nonconvergence

obstruction explain both the usefulness and the limitations of these formulas. The forward resolvent proposition is a secondary classical consequence of the retained source bound. No historical priority or measured speedup is claimed.

8.1 Public primary literature and verification scope

The October 9, 2026 refresh used only public titles and URLs. Domar's publisher record [2] confirms the 1981 weighted-translation paper; its original full proof was not freshly audited. The cached source's included quadratic proof was read. Kérchy's July 4, 2026 preprint [3] concerns operators containing unilateral shifts and a block-triangular similarity model; only its primary abstract and version record were read. It is not used as a proof of the present global source claim.

The Kérchy–Pearcy preprint [4] is marked withdrawn in its March 29, 2025 version, with a reported counterexample to its Theorem 3. Its claimed general relationship between invariant and hyperinvariant subspace problems is therefore not imported. The repository's public formalization-scope page [5] declares a direct-model quasinilpotent operator with a transitive proper strongly closed commutant, alongside a companion construction. This is a scope declaration only: no Lean execution or independent formal proof audit was performed. Neither metadata nor byte identity establishes mathematical correctness.

8.2 Advantages, costs, and future questions

The finite normalization is exact on its stated event, handles both ends with their different boundary indices, and uses exact integer logs before exponentiation. The norm estimates apply to arbitrary strongly measurable H_p inputs, not only simple functions. The lower bound shows that the power cannot be improved by merely refining the upper-bound proof for the same cutoffs.

The cost is the large window needed for a small exceptional-set budget and the restriction $p > 2$ for a uniform vanishing error. Coefficient tables do not discretize the base in operator norm. They give neither exact commutant membership nor classification of closed invariant subspaces. Changing the weights or the adding-machine law requires checking the recurrence, tail bounds, probability law and any classification hypotheses again.

Useful next steps are interval-certified evaluation of the finite tables at a specified input integrability budget, sharper window constants without changing the power, and input-adapted base truncation when additional distributional information about $\|f(x)\|_K$ is available. Whether another approximation class can improve the cost is open within this report; the lower bound is not a complexity lower bound. Any stability theorem for the global hyperinvariant conclusion would require new exact structural control, not just (13) or the resolvent perturbation budget.

9 Read scope and reproducible finite checks

PRE hashes were persisted before the fresh scientific displays. All 13 inventoried public texts were literally displayed in bounded, untruncated chunks: 1,165 unique lines, including all scientific sections, the entire included Domar proof, references, wrapper, figure code and metadata. The three private original chapter bodies total 620 freshly read lines. Their included external figure, referenced chapters and fixed-point result, companion lemma bodies, historical full reference proofs and the rest of the private corpus remain outside the read scope. The final ledger records the actual inclusive display intervals, their unions, exact unread complements within the inventory and PRE/POST SHA-256 values. Complete literal reading is not labelled an independent global theorem certification.

The owned script `M587-checks.py` uses Python integers and fractions for 189 parameter cases with 685 assertions. An assertion may compare an entire specified finite array; entries inside such a comparison

are not inflated into separate cases or assertions. It checks the two signed log formulas, intertwining coefficient identity, exact normalizers and digit invariance for all good residues at $R = 0, 1, 2, 3$, finite extensions to twice the window, escape monotonicity, exact probability bounds, cutoff boundary columns, source power/ratio identities, and Example 3.2. Six separate float64 cases with six assertions diagnose the Gaussian resolvent majorant; these are not interval-certified inequalities. All checks pass. There are no parent or foreign cases in these totals. Finite extensions of a window are diagnostics, not the proof of its infinite complement; that proof is Lemma 2.2.

The contribution score C3 records these source-specific local increments. Applicability L3 remains a separate catalogue property. No claim of a new solution to a global invariant-subspace problem, full-corpus reading, or independent verification of the source's complete transitivity argument is made.

References

[1] OpenAI, *Backward intertwiners and a transitive commutant*, September 27, 2026. Cached manuscript and included scientific modules. Public source directory. Captured-tree identity is separate from proof validity.

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[4] L. Kérchy and C. Pearcy, *Hyperinvariant subspaces of block-triangular operators on Hilbert space*, arXiv:2503.13005, v2, March 29, 2025, withdrawn. Withdrawal metadata and abstract read; not used as a theorem.

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[6] M. Subbarao, original private RamCalBooks, volume 21, *state/domain chapter*, local file `ch01a-state-domain.tex`, lines 1–164, freshly read. Confidential local material; not transmitted online.

[7] M. Subbarao, original private RamCalBooks, volume 21, *spectral-limits chapter*, local file `ch01f-spectral-limits.tex`, lines 1–242, freshly read. Confidential local material; not transmitted online.

[8] M. Subbarao, original private RamCalBooks, volume 21, *weighted-sequence chapter*, local file `ch10d-sequence.tex`, lines 1–214, freshly read. Confidential local material; referenced external proof bodies excluded.